

F1 Podium Finish Prediction Using Race Result Data

Introduction

My project is based on my interest in Formula 1 racing. I will be using data from races spanning from 2019-2024. After the recent race here in Miami, I was curious about the variety of factors that contribute to a successful team season and how historical data could be leveraged to potentially predict future race results. While some more advanced approaches focus on data specific to a vehicle's telemetry, environmental conditions or more granular points of information, I will be focusing on analyzing more general race data to compare and contrast various team's historical performance. The goal of this project will be to predict a podium vs. non-podium finish for a future race provided similar feature data.

Related Work

The GitHub notebook "Formula 1 Race Results 2019–2024" provides a practical example of exploratory data analysis using race result data. It demonstrates how F1 data can be loaded, inspected, cleaned, and visualized to better understand patterns in driver and team performance. Although it is not a formal research paper, it is useful for this project because it works with a similar type of race-level dataset and highlights the importance of preprocessing before applying machine learning methods.

The "Formula 1 Race Results" repository provided a useful template for my project and included an exploratory analysis section that focused on top drivers, top teams, ranking drivers by win count, podium finishes, fastest laps, points by season, along with a variety of other vectors of comparison. The feature engineering section employed one-hot encoding, scikit-learn's simple imputer along with other scikit-learn built in module methods. Training and test split was also 80/20, and the models employed for prediction were Logistic Regression, Random Forest, Gradient Boosting and Extra Trees. The target for these models was also "Podium Finishes"; however this project did not include 2nd place finishes.

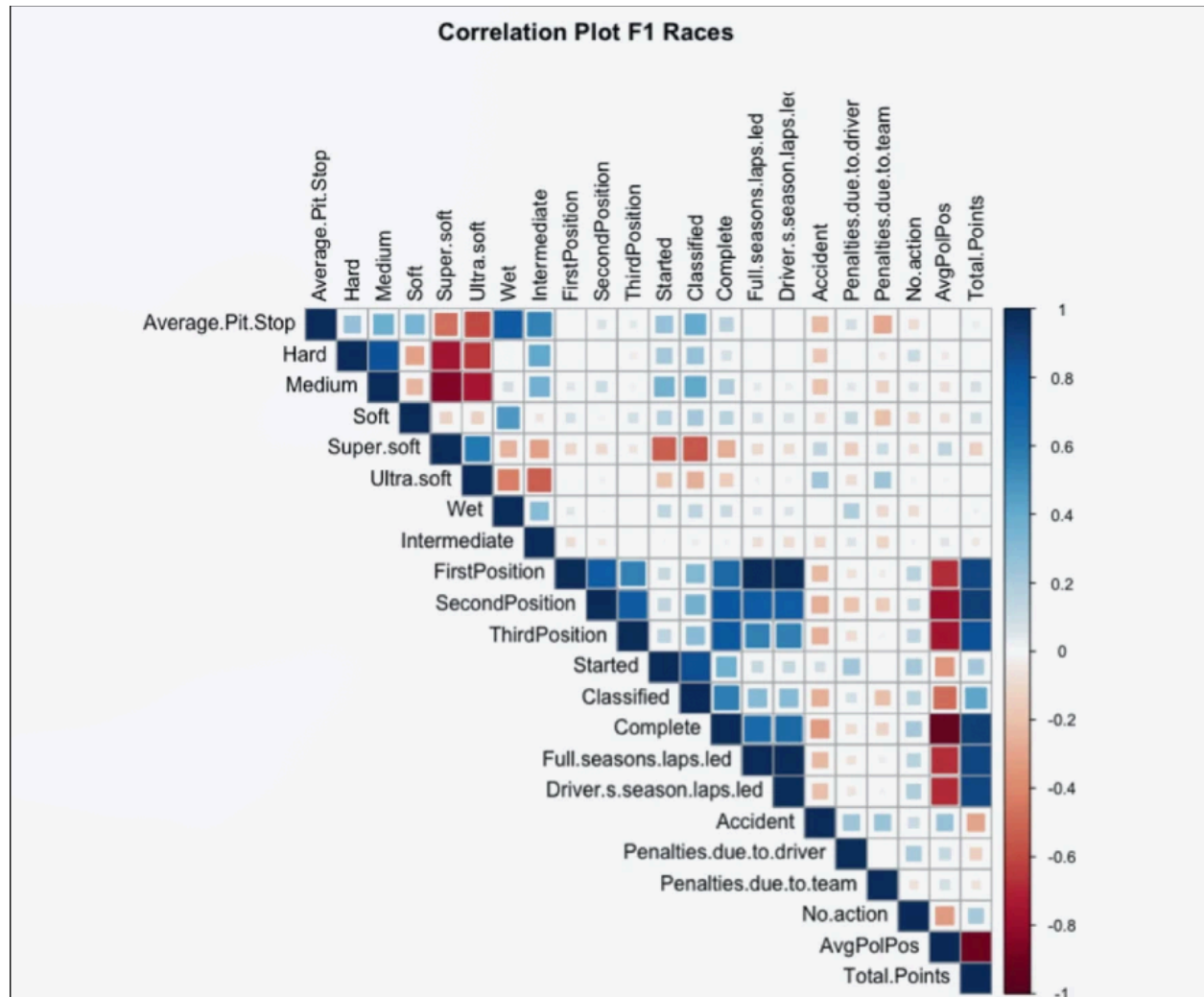
Performance:

Model Comparison						
	Model	Accuracy	Precision	Recall	F1	ROC_AUC
0	Logistic Regression	0.912109	0.705128	0.714286	0.709677	0.944380
2	Gradient Boosting	0.910156	0.701299	0.701299	0.701299	0.937035
1	Random Forest	0.898438	0.676056	0.623377	0.648649	0.929482
3	Extra Trees	0.892578	0.644737	0.636364	0.640523	0.917734

The best ROC AUC (Receiver Operating Characteristic - Area Under the Curve) score for this project was 94.4% from the Logistic Regression model trained with max iterations set to 5000.

Patil et al. (2023) conducted a data-driven analysis of Formula 1 race outcomes using historical racing data. Their work is relevant because it treats Formula 1 as a prediction and analytics problem rather than simply a sports summary problem. Race outcome prediction is challenging because results are influenced by many interacting variables, including driver ability, team performance, race conditions, and starting position. This supports my decision to simplify the task into a binary podium classification problem instead of attempting to predict exact finishing order.

In the Springer Nature research article “A Data-Driven Analysis of Formula 1 Car Races Outcome”, data from F1 races between 2015 and 2019 are explored and a Principal Component Analysis (PCA) is performed to identify the most impactful attributes out of a larger space of 21 input variables. The article breaks their research into four main categories for strategy that include: preparation, practice, qualifying and racing.



As the article mentions, based on the results of the correlation plot (above), pole position is highly correlated with race outcome. The article notes “We observe that the average pole position is strongly negatively correlated with the first, second and third position.”, and continues with an observation that team penalties “appear to be related to the usage of soft tyres and hyper soft tyres”.

The Catapult article discusses how data analysis is used in modern Formula 1 to improve performance, strategy, and decision-making. While the article is industry-focused rather than academic, it provides useful context for why F1 is an appropriate domain for machine learning. It shows that race teams rely heavily on data to understand performance and make decisions. My project applies this idea on a smaller scale by using publicly available race result data to predict podium finishes.

In this analysis, the real-world integration of Catapult's solutions "Racewatch" and "Circuit Manager" are discussed in depth. The main advantages covered for integration of these modern tools is in the categories of vehicle performance optimization through telemetry data, predictive analytics for improving strategy, as well as driver analytics (how do drivers behave under different environment conditions)

The article continues to discuss how the sport is evolving through the leveraging of big-data analytics, AI and machine learning integration, Virtual and Augmented Reality technologies and Internet of Things (IoT) telemetry data. Hardware provides more "granular" data on areas such as tire wear, aerodynamics and engine output, while AI and Machine Learning facilitate strategy improvements.

Data

The dataset provides a variety of numerical and categorical features across **15** columns with a total of **2559** unique rows.

- My **target** variable will be a binary classification of a podium finish (1st, 2nd or 3rd place) vs. a non-podium finish.
- I've sourced my data from Kaggle, and the dataset was downloaded in csv format for ingestion.
- Initial inspection of the data indicates zero duplicated and a significant number of missing values for a small group of the 15 feature columns (ex. The "Fastest Lap" column contains **970** empty values)
- I'll need to exclude the racer number column from my model training, as it is primarily a unique identifier and not useful for analysis.
- I will drop the "+1" column from my model training to prevent data leakage from the training into the test set.
- Data Leakage is a potential concern for some of the features whose values are conditional on the target/dependent variable (ex. "+1" column is a bonus for top performers). Columns such as "Total Time/Gap/Retirement" and "Time/Retired" need to be removed or adjusted due to their conditional linkage to the outcome variable (podium finish). Because the target variable is derived from the final race position, I will avoid all race outcome fields.

Implementation

- *Libraries used*
- *Pipeline*
- *Data preprocessing steps*
- *Feature extraction*
- *Classifiers, regressors*

- *Performance measurement techniques (accuracy, precision, F1 scores)*
- Modeling starts with a K-Nearest Neighbors (KNN) classifier initially and compared against results from Logistic Regression along with XGBoost.
- My training/test split will be 80/20 and I will apply this consistently in subsequent models
- If 2026 results become available, I will use that data for external validation, otherwise I'll use results from 2025.
- I will implement KNN and Logistic Regression using the scikit-learn library and XGBoost library for their XGBoost model.
- For metrics, I will be using precision, recall, F1 score (harmonic mean of precision and recall) and ROC-AUC (Receiver Operating Characteristic and Area Under Curve) to evaluate and compare performance of the different models. If time permits, I will add PR-AUC (Precision-Recall Area Under Curve) to provide a more conservative estimate of performance.

Results

- *Present findings alongside comparison of related works*
- *Visual materials (graphs, confusion matrices, ROC AUC graph)*

Discussion

- What went well
- What didn't go so well
- System performance?

Future Work

References

1. Patil, A., Jain, N., Rahul Agrahari, Murhaf Hossari, Orlandi, F., & Dev, S. (2023). A Data-Driven Analysis of Formula 1 Car Races Outcome [Review of *A Data-Driven Analysis of Formula 1 Car Races Outcome*]. *Springer Nature*, 134–146.
https://doi.org/10.1007/978-3-031-26438-2_11
2. Ambler, W. (2024, February 13). F1 Data Analysis & Technology: How Data is Transforming Race Performance. Catapult.
<https://www.catapult.com/blog/f1-data-analysis-transforming-performance>
3. Sonawane, L. (2026). formula-1-race-results-2019-24-94.ipynb (Commit 79b74be) [Jupyter Notebook]. GitHub.

<https://github.com/lalitssonawane/100-days-of-data-science-projects/blob/79b74bef5684f2bec33d6ee6c8d4e46c9ca3f2a5/formula-1-race-results-2019-24-94.ipynb>